

Definition

1. coenzyme: non-protein organic compound that binds to apoenzyme to form holoenzyme. It is responsible for catalyse and serve as carries of specific atoms. 催化 + 载体
2. apoenzyme: an inactive enzyme that requires coenzyme to become active
3. cofactor: non-protein compound essential for enzyme activity
4. prosthetic group: A prosthetic group is a non-protein, tightly-bound component of a protein, often an enzyme.
5. Intron: non-coding regions removed from spliceosome
6. Extron: coding regions removed from spliceosome
7. Codon: adjacent 3 nucleotide
8. Anticodon: consists of 3 adjacent base that match with code in mRNA
9. Active side of enzyme: where substrate molecules bind and undergo chemical reaction. 活性位
10. Non-competitive inhibition: inhibitor bind to allosteric site
11. Competitive inhibition: inhibitor bind to active site
12. Enzyme kinetics: at high substrate concentrations, reaction velocity of enzyme reached to maximum.
13. Missen constant: the concentration of substrate when velocity reached to half of maximum.
14. Transferase: catalyse the transfer between functional groups.
15. Telomerase: lengthen telomers by adding repetitive sequences.
16. Central dogma: information flow: DNA $\rightarrow$ RNA $\rightarrow$ protein
17. Core enzyme: $\alpha 2\beta\beta'$ polymerize
18. Holoenzyme: $\alpha 2\beta\beta'$ complete enzyme
19. Phosphodiester bond: covalent linkage between 2 nucleotide when the 2 hydroxyl group from each bond to the same phosphate group
20. Transcription 3 stages: initiation, elongation, termination
30分

21. 5 stages of mRNA synthesis in eukaryotic

① capping: 5' cap is added to the 5' end of newly synthesized mRNA.

② Splicing: ^{RNA}excision of introns & formation of final mRNA molecule

③ Spliceosome 31. a key metabolic pathway that occurs in the mito

④ Cleave essential for energy production, in aerobic

⑤ Editing { 3 NADH, 1 FADH₂, 1 ATP, 2 CO₂ } ① Acetyl-CoA + oxaloacetate → citrate
② citrate → release CO₂ & transfer electrons to NADH,
③ regenerate oxa, continue

22. Telomeres: the protective cap that locate at the end of chromosomes (middle)

23. Ketone bodies: intermediate product from fatty acid oxidation, include acetoacetic acid, β -hydroxybutyric acid and acetone.

24. 米氏常数: It is a key parameter in enzyme kinetics that describes the affinity of an enzyme for its substrate.

25. Beriberi is caused by a deficiency of VB₁, an essential nutrient required for carbohydrate metabolism & proper nerve, muscle, heart function.

26. DNA → RNA → protein

In sickle cell anemia, a mutation in the HBB gene causes a single amino acid change in the hemoglobin protein. This alters the protein's shape, leading to sickle-shaped red blood cells. This condition is directly linked to the altered protein production as per the central dogma.

27. tRNA: ① an RNA molecule helps decode mRNA → a specific sequence of amino acids to the ribosome

② peptide bond formation ③ anticodon recognition

28. mRNA: It is a type of RNA that carries genetic instructions from DNA to the ribosomes for protein synthesis.

29. It's a process in which a phos direct transfer of a phosphate group from a substrate to ADP, forming ATP.

30. the process where glucose is broken down into pyruvate, producing ATP & NADH, and occurs in the cytoplasm.

Chapter 11 Protein Synthesis

1. Protein translation occurs in ribosome.
2. The first amino acid incorporated into a polypeptide chain in eukaryotes is typically Methionine. 甲硫氨酸
3. The sequence of three nucleotides on mRNA that codes for an amino acid is called codon. 三联体密码子
4. tRNA delivers amino acids to the ribosome during translation.
5. The termination of protein synthesis occurs when the ribosome encounters a stop codon.

Chapter 15 Vitamin Metabolism

1. Vitamin B complex

① B₁ (Thiamine)

- (1) Dry Beriberi: peripheral neuropathy, muscle weakness, numbness, atrophy
- (2) Wet Beriberi: heart failure, edema, and rapid heartbeat
- (3) Infantile 婴儿 Beriberi: respiratory distress, heart failure
- (4) Wernicke's Encephalopathy: Confusion, ataxia

② B₂ (Riboflavin)

- (1) Cheilitis 口角炎: Cracked 干裂的 lips, angular stomatitis 口角炎, Swollen tongue
- (2) Ocular Symptoms: Conjunctival hyperemia, photophobia, burning sensation
- (3) Skin Symptoms: Seborrheic dermatitis
- (4) Neurological symptoms: Fatigue and weakness

③ B₉ (Folic Acid)

- (1) Hematological Symptoms: Megaloblastic anemia
- (2) Pregnancy Issues: Neural tube defects in the fetus.
- (3) Others: Fatigue, mouth ulcers

④ B₁₂ (Cobalamin)

- (1) Hematological 同上
- (2) Neurological Symptoms: Numbness, sensory loss, balance issues
- (3) Cognitive Symptoms: Confusion, memory loss

2. Vitamin A

Night blindness

Xerophthalmia 干眼症

Growth retardation 缓慢

3. Vitamin C

Scurvy: swollen gums 牙龈, slow wound healing

4. Vitamin D

Rickets in children: bone deformities

Osteomalacia in adults: bone pain and fractures

5. Vitamin E

Peripheral neuropathy

Muscle weakness

Retinopathy

6. Vitamin K

Bleeding disorders: easy bruising 瘀青, prolonged bleeding

Hemorrhagic disease of the newborn

Oxygen acts as the final electron acceptor in the electron acceptor in the electrons ~~and~~ and protons to form water, enabling continuous ATP production.

help digest fats, which are then broken down into fatty acids that can enter the CAC for energy production